Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 46 Introduction to Moment of Inertia

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

CLASS 46 Introduction to Moment of Inertia

1 CLASS 46 Introduction to Moment of Inertia

Study focus: CLASS 46 Introduction to Moment of Inertia. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Moment of inertia is an object's resistance to bending at the rotation, and it is also called as second moment of an area.

Key details

- The concept of first moment of an area we discussed in locating the centroid of a lamina.
- It's nothing but second moment of an area.
- That's why moment of inertia I will be equal to distance square d square into area A.
- And it is also.
- A purely a mathematical property of an area and itself doesn't have any physical significance.
- Rectangular moment of inertia is nothing but moment of inertia phenamina about any two mutually perpendicular axes.
- Because X and Y we know they'll be always mutually perpendicular to each other.
- Then moment of inertia about an axis which is perpendicular to the plane of an area or a lamina that we call it as a.
- Polar moment of inertia.
- As we know, z axis is always perpendicular to xy plane or a plane of a lamina.
- You can see here.
- It is in the XY plane.
- about the x axis then about the y axis.
- Let the area of this element let it be d.
- I want to calculate.
- Since area is very small, the centroid of this area will be the coordinates of the centroid are going to be x, y.

- Distance is y, so that we know.
- This is moment of inusia.
- Since we are talking about the smaller element, it.
- We go, we are going to take it as DIX.
- This is this stands for moment of inertia of the elemental area about x-axis.
- Again it is distance square into area.
- Distance here is x.
- Once we know moment of inertia of the element about x and about y axis.
- That is ix is equals to integration.
- integral of y square into d a and iy is integral of x square into d a. i x stands for inertia of the lamina about x axis iy stands for inertia of the lamina about y axis since x and y are perpendicular to each other i x and i y.
- Why we call them as rectangular moment of inertia.
- As I said before, moment of inertia about z axis we call them as polar moment of inertia.
- Since there is perpendicular to the plane of the lamina.
- Nothing but perpendicular to XY plane, we can't see that.
- Just remember that is passing through point O.
- Then how to find inertia about the Z axis passing through O?
- We know moment of inertia is distance square into area.
- This is the moment of the smaller element about the z axis.
- Next, I want to find the moment of inertia of this entire lamina about.
- Z-axis, so just differentiate integrate that.
- This is called as polar moment of inertia.

Example or application from the lecture

- Example like X and Y axis.
- We consider moment of inertia about z axis as the polar moment of inertia.
- To explain this, I am going to consider this figure.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

There is a relationship between polar moment of inertia and rectangular moment of inertia according to perpendicular axis theorem.

Key details

- According to perpendicular axis theorem, moment of inertia about a z-axis, which is perpendicular to the plane of the lamina, will be always equals to.
- Sum of the moment of inertia about the two mutually perpendicular axes passing through the same point O and lying in the same plane of the lamina.
- Nothing but IZ is equals to Ix plus I.
- The condition is.
- X, Y, Z should be passing to the same point.

- If you look at this, there is a right angle triangle.
- Once you substitute here, i z will be equals to integral of x square into d a plus integral of y square into d a.
- From here we know integral.
- integral of x squaring to d is i y and integral of y squaring to d a is i x so iz will be equals to i x plus i y.
- Once we know the rectangular moment of inertia about any point, like O R it can be anything, then we can find polar moment of inertia about the same point just by adding Ix and Iy.
- We are going to get this IZ.
- An element whose coordinate is negative contributes as much as the moment of inertia as does an equal element with a positive coordinate of the same magnitude.
- Since these two are identical, assume.
- The centroid will be some area.
- Since this point centroid is on the right side of the axis of symmetry, it will be positive.
- But since we are talking about the distance square, even though it is on the left side of the axis, even though it is minus x, distance square of the minus 6 will be positive.
- It's a continuation of the same thing.
- Whereas the first moment of area can be positive, negative, or it can be zero, whereas the second moment of an area will be always positive.
- As I said before.
- First moment of an area, we use this concept in locating the centroid.
- Because first moment of an area is area into distance, if the x coordinate is negative, then.
- The first moment of area can be negative.
- Few times it will be zero.
- When the coordinate is positive, first moment of area can be positive.
- Since here it is distance square, so it doesn't matter whether that is positive or negative.
- Next is radius of guidation.
- The total area of the lamina let it be A.
- Then Ix stands for moment of inertia of the lamina about x-axis.
- Iy stands for moment of inertia of a lamina above y-axis.
- X bar and y bar are the coordinates of the centroid.
- This is what we know in this particular case.
- We know area, we know Ix, Iy, X bar, and Y bar.
- We are going to visualize this area as a narrow rectangular strip.
- Just area remains the same area as.
- The same area here, and the moment of inertia of this narrow rectangular strip, which is parallel to x-axis, that is the same as that of the case one.
- The inertia in this case.
- Is also Ix.
- Then how to find kx?

Example or application from the lecture

- To prove that, I'm going to consider here.

- Next, say there is a lamina like this, and this is the y axis given in the cushion.
- To explain this, I am going to consider this rectangle.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

We know from the definition of moment of inertia, moment of inertia is distance square into area.

Key details

- Since we know Ix from here and a from this case, now kx can be written as under root of ix by area.
- Area as a narrow rectangular strip which is parallel to y axis.
- Then inertia of this narrow strip about y-axis is also the same as that of.
- This case i y.
- Ky will be equal to under root of iy by total area A.
- And in this case also, momentum inertia of this area about.
- The point about the z-axis passing through O will be the same as that of the moment of finesse about the z axis passing through O in this case.
- Ky is under root of IY by A, Kz is equals to under root of IZ by area.
- When we know the value of moment of inertia about x-axis, then and we know total area, then we can find radius of variation about x-axis.
- And it is a measure of distribution of the area from the axis in the cushion.
- And as we know here, I z is equals to Ix plus I.
- I z we know kz square into area, that will be equals to kx square into area plus ky square into area.
- Since area is.
- Thus, the square of the radius of guaration about the polar axis equals to the sum of the squares of the radii of variation about the two corresponding regular rectangular axis.
- It's also called as transfer of axis.
- According to parallel axis theorem, moment of inertia about any axis will be always equal to sum of the moment of inertia about parallel centroidal axis plus product of the area and square of the distance between the parallel.
- This is the statement of the.
- Parallel axis theorem mark you can write it as dh.
- That is the distance between the two x axis.
- Like the area of this lamina let it be capital A.
- A is the area of the lamina.
- This is x-axis and this is.
- Let C be the centroid of this lamina.
- Then let it be at a distance of dx with respect to.
- We know how to find DX.

Example or application from the lecture

- If you consider moment of inertia about any horizontal axis, so I'm going to write it as ix will be always equals to moment of inertia about centroidal x axis.
- To prove this, I am going to consider there is a lamina like this.
- Next, consider this is in the xy plane like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next, I am going to consider one smaller elemental area within this lamina.

Key details

- The shaded region, let it be the smaller elemental area, let d be the area of this element.
- Let it be at a distance of 10 with respect to.
- The reference line here, the reference line is centroidal x-axis, so let it be y naught.
- Smaller element above the x axis.
- What is moment of inertia?
- Moment of inertia is second moment of an area, that is distance square into area.
- Distance from ear to ear is y naught plus dx.
- Whole square into area is dA.
- This is the moment of inertia of the smaller element about the x-axis.
- It's going to be a square.
- Y not square into DA.
- What is this?
- It is dis it is in the form of distance square into area.
- What is distance square into area?
- Distance square into area is second moment of an area.
- Nothing but moment of inertia.
- Since the distances with respect to centroidal x axis and area is here, it is moment of inertia bar centroidal x axis.
- Second term is integral of dx square into dm.
- This is constant, dx is constant.
- What is d?
- Integral of d a integral of d a is total area.
- The last term is two into integral of y naug into dx into d.
- This will be equal to here.
- Dx is constant.
- Here I am going to divide it by total area and multiply with total area.
- It doesn't make any difference.
- Since the central is on the reference line, this will be equal to 0.
- This entire term will be 0.
- So we are going to get IX as Ix is equal to first term integral of y naught square into d is i bar x.
- I bar X plus this is dx square into A and this is 0.

- Nothing but moment of inertia about any axis will be always equal to.
- Sum of the moment of inertia about parallel centroidal axis plus product of the area and square of the distance between the two parallel axes.
- Here you have to remember two points to apply the parallel axis theorem.
- The first condition is the axis.
- And the centroidal axis both should be parallel to each other.
- And to use this formula, one of the axes should be the centroidal axis.
- Then only we can apply parallel axis theorem to find the moment of inertia above any axis.
- that's it for today's class.

Example or application from the lecture

- Next, consider each term.
- First, I am going to consider the term integral of.
- Next consider the second term.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 46 Introduction to Moment of Inertia transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 46 Introduction to Moment of Inertia transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 46 Introduction to Moment of Inertia transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
CLASS 46 Introduction to Moment of Inertia transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00003	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00005	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

C Revision and retrieval practice